

Question		Solution	Marks
1	(a)	<u>Calculations if $g_1 = 8.80 \text{ m s}^{-2}$ rejected as anomalous:</u> $\langle g_1 \rangle - 9.81 = 10.52 - 9.81 = 0.71$ $\langle g_2 \rangle - 9.81 = 9.78 - 9.81 = -0.03$ Since $\langle g_1 \rangle$ is further away from the accepted g value, results for group 1 have larger systematic errors.	M1 A1
	(b)	Range of $g_1 = 10.56 - 10.48 = 0.08$ Range of $g_2 = 10.40 - 9.16 = 1.24$ Since values for g_2 has a larger range compared to values for g_1, results for group 2 have larger random errors.	M1 A1
		<u>Calculations if all data used:</u> $\langle g_1 \rangle - 9.81 = 10.23 - 9.81 = 0.42$ $\langle g_2 \rangle - 9.81 = 9.78 - 9.81 = -0.03$ Since $\langle g_1 \rangle$ is further away from the accepted g value, results for group 1 have larger systematic errors.	max 3 marks (anomaly not accounted)
		Range of $g_1 = 10.56 - 8.80 = 1.76$ Range of $g_2 = 10.40 - 9.16 = 1.24$ Since values for g_1 has a larger range compared to values for g_2, results for group 1 have larger random errors.	

Question			Solution	Marks
2	(a)		The rate of change of momentum of a body is proportional to the resultant	B1